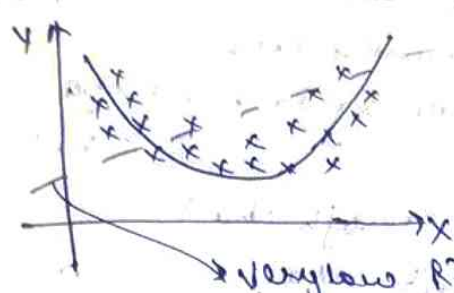


Polynomial Regression:



$$y = \beta_0 + \beta_1 x$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

(LR) (multiple)

degree $\rightarrow 2$
hyperparameter.

x_1	y		
35	100	(1, 2)	
x_1^0	x_1^1	x_1^2	y
1	35	35 ²	(1, 4)

(for low degree)

underfit

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2$$

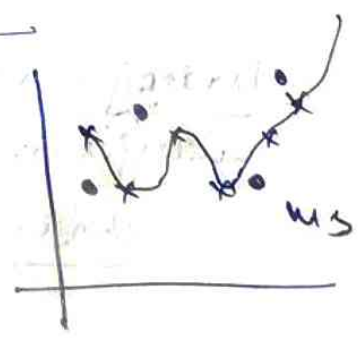
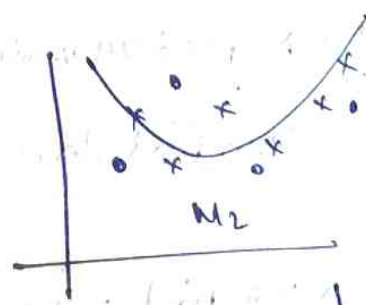
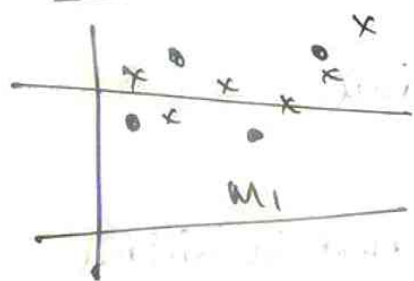
for high degree
overfit)

$$x_1 \quad x_2 \quad | \quad y \quad \beta \quad D=2$$

$$y = \beta_0 + (\beta_1 x_1 + \beta_2 x_1^2) + (\beta_3 x_2 + \beta_4 x_2^2)$$

Polynomial Regression is a type of LR as in LR we check relationship b/w y and $\text{way} (\beta_1, \beta_2, \dots)$
As it is still linear, it is a type of LR

Bias Variance Tradeoff:



$x \rightarrow$ Training set
 $\bullet \rightarrow$ Test set

Bias → The inability of a ML model to truly capture the relationship in the training data.

(Samajh nahi para training data me kya hai)

M_3 → low bias (Ache se samajh rha)

M_1 → high bias (Nhi samajh rha)

Variance → difference in fits on different datasets

error is to
$$\text{variance} = |(\text{error}_{\text{training}} - \text{error}_{\text{test}})|$$

$\left\{ \begin{array}{l} M_1, M_2 \rightarrow \text{low variance} \\ M_3 \rightarrow \text{high variance} \end{array} \right.$

error train = 0
error test dominates.)

• Overfitting: When a ML model works exceptionally well on training data but not on test data. (M_3)

• underfitting: Model that is not able to capture training data properly. works bad on test data. (M_1, M_2)

For A good ML model

Low Bias, low variance

Tradeoff → we find a model that is neither underfit/overfit having LB, LV. (M_2 is chosen among M_1, M_3)

3 methods for Tradeoffs:

- 1) Regularization
- 2) Bagging
- 3) Boosting